

Research Methodology in CS (CSEG3060_3)

Lecture 6: New Developments in IPR - Administration of Patent System, recent reforms, and global harmonization (TRIPS, PCT)

Unit 6: Patent Rights and New Developments in IPR • Mapped Course Outcome: CO4

Lecture 6: New Developments in IPR — Administration of the Patent System, Recent Reforms, and Global Harmonization (TRIPS, PCT)

Course: Research Methodology in Computer Science (CSEG3060_3) **Program:** B.Tech (Computer Science) — Semester V **Unit:** 6 — Patent Rights and New Developments in IPR **Mapped Course Outcome:** CO4 — Analyze contemporary developments in Intellectual Property Rights with specific reference to patent administration, global harmonization mechanisms, and their implications for research and innovation.

1. Lecture Overview and Learning Objectives

The intellectual property landscape has transformed dramatically since the late twentieth century. What began largely as a patchwork of national patent laws has, over four decades, evolved into an internationally coordinated framework anchored by treaties such as the **Paris Convention (1883)**, the **Patent Cooperation Treaty (PCT, 1970)**, and most decisively, the **Trade-Related Aspects of Intellectual Property Rights (TRIPS) Agreement (1994)**. For computer science researchers, software developers, and technology entrepreneurs, the administration of these systems is no longer an optional concern — it is a strategic determinant of competitive advantage.

By the end of this lecture, students should be able to:

- Describe the institutional architecture of the global patent system, distinguishing between national, regional, and international administrative bodies.
- Explain the principal features and procedural mechanics of the TRIPS Agreement and the Patent Cooperation Treaty.
- Identify and critically assess major recent reforms in patent administration, particularly as they relate to software and computer-related inventions.
- Evaluate the impact of global harmonization on innovation strategy in computer science research.

- Apply this knowledge to interpret contemporary case law and policy debates.
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2. Key Concepts Introduced

- **Administration of the Patent System** — the institutional and procedural machinery through which patent rights are granted, maintained, and enforced.
 - **Patent Office** — the governmental authority responsible for examining and granting patents (e.g., USPTO, EPO, IPO, JPO, CNIPA).
 - **Patent Prosecution** — the back-and-forth process between applicant and patent office leading to grant or refusal.
 - **TRIPS Agreement** — the WTO-administered treaty establishing minimum IP standards across member states.
 - **Patent Cooperation Treaty (PCT)** — a WIPO-administered international filing and search mechanism.
 - **Global Harmonization** — the convergence of national patent laws toward shared standards, principles, and procedures.
 - **Software-Related Inventions** — innovations in computer-implemented subject matter, governed by increasingly harmonized but still variable national rules.
 - **Budapest Treaty, PLT, and PCT Reform Packages** — supporting international instruments.
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3. Administration of the Patent System

3.1 The Institutional Architecture

Modern patent administration operates at three interconnected tiers.

Tier 1 — National Patent Offices. Each sovereign state maintains a national or regional office that examines applications under domestic law. The most consequential offices are:

- **United States Patent and Trademark Office (USPTO)** — the largest national office by volume.
- **European Patent Office (EPO)** — a supranational office that examines applications under the European Patent Convention (EPC) for contracting states.
- **Japan Patent Office (JPO).**
- **China National Intellectual Property Administration (CNIPA)** — the fastest-growing filer of patents globally.
- **Indian Patent Office (IPO)** — operating through its four regional offices at Delhi, Mumbai, Chennai, and Kolkata under the Patents Act, 1970.

- **Korean Intellectual Property Office (KIPO).**

Tier 2 — Regional Patent Systems. Regional offices such as the **Eurasian Patent Organization (EAPO)**, the **African Regional Intellectual Property Organization (ARIPO)**, the **African Intellectual Property Organization (OAPI)**, and the **European Patent Office (acting as a regional grant body)** consolidate filing procedures for groups of countries.

Tier 3 — International Coordination Bodies. The **World Intellectual Property Organization (WIPO)**, headquartered in Geneva, administers the PCT, the Madrid System for marks, and the Hague System for designs. WIPO functions as a treaty secretariat and a normative standard-setter rather than a granting authority.

3.2 The Patent Prosecution Process

Patent prosecution follows a standardized workflow, even though details vary across jurisdictions:

1. **Filing** — submission of the application (specification, claims, abstract, drawings) to the office of first filing (OFF).
2. **Formalities Examination** — verification of administrative requirements.
3. **Search** — identification of prior art; produces a search report.
4. **Substantive Examination** — assessment against patentability criteria (novelty, inventive step/non-obviousness, industrial applicability, sufficiency of disclosure).
5. **Office Actions and Responses** — exchanges between examiner and applicant, often involving claim amendments.
6. **Grant or Refusal** — culminating decision; refused applications may be appealed.

The standard patentability criteria, codified across most jurisdictions and now in TRIPS, can be expressed as a logical conjunction:

$$\text{Patentability} = N \wedge IS \wedge IA \wedge SD$$

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where NN denotes novelty, $ISIS$ denotes inventive step (or non-obviousness), $IAIA$ denotes industrial applicability (or utility), and $SDSD$ denotes sufficient disclosure. Each condition must be independently satisfied.

3.3 Post-Grant Administration

Beyond grant, the patent system requires ongoing administration:

- **Renewal/Annuity Payments** — periodic fees to maintain the patent in force, typically escalating over the 20-year term.

- **Opposition Procedures** — third parties may challenge the validity of granted patents (administered through offices such as the EPO under Article 99 EPC or the Indian Patent Office under Section 25 of the Patents Act, 1970).
 - **Invalidation and Revocation** — judicial or administrative processes for canceling patents.
 - **Licensing Registers** — public records of contractual patent exploitation.
 - **Patent Information and Databases** — systematized repositories described in Section 5 below.
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4. Recent Reforms in Patent Administration

Patent administration has undergone continuous reform since the early 2000s. Three themes dominate: (a) acceleration of prosecution, (b) accommodation of new technologies, and (c) digital transformation.

4.1 Acceleration and Quality Reforms

- **USPTO Track One (Prioritized Examination)**: introduced under the **Leahy-Smith America Invents Act (AIA), 2011**, providing expedited examination for an additional fee, with a final disposition target of 12 months.
- **Patent Prosecution Highway (PPH) Networks**: bilateral agreements allowing participating offices to fast-track applications relying on work-product from partner offices, reducing duplication and search costs.
- **EPO "Raising the Bar" Initiative (post-2010)**: tightened examination standards, expanded use of supplementary European search reports, and increased reliance on examiner-added prior art.

4.2 Software-Related Subject Matter

Software remains the most contested category. Historically, computer programs as such were excluded from patentability in jurisdictions such as India (Section 3(k) of the Patents Act, 1970, before amendment) and the European Union (Article 52 EPC). However, technical applications of software are routinely patentable.

Recent reforms include:

- **USPTO Revised Patent Subject Matter Eligibility Guidance (2019 and updates through 2024)**: clarifies the application of the two-step Alice/Mayo framework for abstract ideas, with specific examples in artificial intelligence and computer-implemented inventions.
- **EPO Compendium of Case Law updates**: continuous refinement of the "technical character" requirement for software claims.

- **Indian Patent Office Guidelines (2018 and 2024 drafts):** clarified that computer-related inventions (CRIs) may be patentable when they demonstrate a "technical effect" or "technical advancement."

4.3 Reforms Driven by Emerging Technologies

The proliferation of artificial intelligence, biotechnology, and blockchain has generated a wave of administrative reforms:

- **AI-Assisted Prior Art Search:** major offices including CNIPA and JPO have deployed machine-learning tools to assist examiners in source-document identification.
- **Deposit of Biological Materials:** the Budapest Treaty on the International Recognition of the Deposit of Microorganisms (1977, amended multiple times) governs the procedural requirements.
- **Standard-Essential Patents (SEPs) and FRAND Licensing:** the Indian Department of Promotion of Industry and Internal Trade issued a draft **Standard Essential Patents Policy** in 2021; the European Commission has advanced FRAND licensing frameworks.
- **AI as Inventor:** the DABUS cases (Thaler v. Comptroller, EPO Boards of Appeal, 2021–2022; Thaler v. Vidal, US Federal Circuit, 2022) rejected artificial intelligence systems as inventors, prompting academic debate and continuing policy analysis.

4.4 Digital Transformation and Pandemic-Era Adaptation

The COVID-19 crisis accelerated digital modernization:

- WIPO expanded **ePCT** filing and electronic priority document exchange via the WIPO Digital Access Service (DAS).
 - The USPTO implemented fully electronic prosecution and launched the **Patent Center** portal replacing the legacy PAIR system.
 - The EPO finalized its **Opposition Oral Proceedings by Videoconference** framework (2021).
 - The Indian Patent Office extended timelines and accepted electronic filing under the Patents (Amendment) Rules, 2020, and continues to extend deadlines through the Patents (Amendment) Rules, 2024.
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5. Global Harmonization

5.1 Conceptual Foundations

Global harmonization in patent law refers to the progressive alignment of substantive standards, procedural norms, and enforcement mechanisms across jurisdictions. The goal is to reduce transaction costs for inventors operating internationally, minimize forum-shopping, and enhance legal certainty.

The fundamental tension in harmonization is captured by:

$$H_{\text{gain}} = \sum_{i=1}^n \Delta C_i - D$$

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where H_{gain} is the net harmonization benefit, ΔC_i is the reduction in compliance cost for country i , and D is the policy divergence cost (the loss from abandoning divergent national preferences). Harmonization is pursued only when $H_{\text{gain}} > 0$.

5.2 TRIPS — The Trade-Related Aspects of Intellectual Property Rights

5.2.1 Genesis

TRIPS emerged from the **Uruguay Round (1986–1994)** of the General Agreement on Tariffs and Trade (GATT), coming into force on **1 January 1995** as Annex 1C of the **Marrakesh Agreement Establishing the World Trade Organization**. It became enforceable not as standalone IP legislation but as a multilateral trade instrument; non-compliance triggers WTO dispute settlement.

5.2.2 Substantive Provisions (Articles 27–34)

Article 27 mandates that patents shall be available for **any inventions**, whether products or processes, in **all fields of technology**, provided they are **new, involve an inventive step, and are capable of industrial application**. This principle of non-discrimination across technology fields is foundational to TRIPS and is consequential for software patents.

Other key provisions:

- **Article 31** — conditions for compulsory licensing.
- **Article 33** — patent term of **twenty years** counted from the filing date.
- **Article 34** — reversal of burden of proof in infringement of process patents.
- **Article 7 and 8** — objectives and principles, balancing rights with public interest.

5.2.3 Flexibilities and Critical Reactions

TRIPS permits significant flexibility:

- **Compulsory licensing** under specified conditions.
- **Parallel imports** governed under "exhaustion" rules.
- **Bolar exemption** allowing generic manufacturers to use patented inventions for regulatory approval.
- **Transition periods** for developing countries.

Critiques include the *Declaration on the TRIPS Agreement and Public Health* (Doha, 2001) and the Article 31bis amendment (2005) allowing export of pharmaceuticals manufactured under compulsory license.

5.2.4 TRIPS-Plus Provisions

Through bilateral and regional trade agreements such as USMCA, the EU-Vietnam FTA, and various EFTA-EPAs, many countries have accepted **TRIPS-plus** obligations — extended patent terms, data exclusivity, patent linkage mechanisms, and restrictions on compulsory licensing — that exceed minimum TRIPS standards.

5.3 The Patent Cooperation Treaty (PCT)

5.3.1 Structural Framework

Administered by WIPO, the PCT currently has **over 150 contracting states**. It establishes a unified international filing procedure that does not grant international patents; rather, it centralizes the initial stages (filing, international search, optional preliminary examination) and defers national-phase entry by up to **30/31 months** from the priority date.

5.3.2 Procedure Summarized

1. **International Application** filed under PCT Article 8, designating contracting states.
2. **International Search** by an authorized International Searching Authority (ISA) — major ISAs include the EPO, JPO, KIPO, CNIPA, and the Indian Patent Office (acting since 2013).
3. **Written Opinion of the International Searching Authority** — first non-binding assessment of patentability.
4. **International Publication** at 18 months.
5. **Optional International Preliminary Examination** under Chapter II PCT.
6. **National Phase Entry** — application proceeds in each designated state under domestic law.

5.3.3 Strategic Significance

For computer science inventors, PCT offers:

- **Deferred cost** — national fees and translations postponed.
- **Search-quality reassurance** — international search report informs strategy.
- **Marketing window** — 18-month publication often suffices for venture or licensing negotiations.
- **Procedural protection** — the international filing date is recognized as a regular national filing date in designated states (Article 11(3)).

5.4 Supporting International Instruments

- **Paris Convention for the Protection of Industrial Property (1883)**: establishes the right of priority (Article 4) and national treatment (Article 2).
 - **Budapest Treaty (1977)**: governs deposit of microorganisms.
 - **Patent Law Treaty (PLT, 2000)**: harmonizes formal requirements.
 - **Vienna Agreement (1973)**: standardizes patent classification; increasingly superseded by the **International Patent Classification (IPC)** and **Cooperative Patent Classification (CPC)**.
 - **Hague Agreement (1960)** for designs and **Madrid Protocol (1989)** for marks — adjacent to patent administration.
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6. Patent Information and Databases (Brief Discussion)

The lecture topic emphasizes administration; accordingly, the support infrastructure merits brief treatment.

- **WIPO PATENTSCOPE** — access to PCT international applications and national collections.
- **Espacenet** (EPO) — over 130 million documents with translation tools.
- **USPTO Open Data APIs (Patent Full-Text and Image Database, PatFT/AppFT)**.
- **Indian Patent Database (IP India)** with searchable full-text retrieval from 1998 onwards.
- **Google Patents** — aggregated search layer with non-patent literature (NPL) integration.
- **Lens.org** — open patent and scholarly literature analytics with PatCite-style citation graphs.

For computer science researchers, these databases are essential for **freedom-to-operate (FTO)** analysis before publishing open-source code, releasing software products, or filing competing applications.

7. Case Studies

7.1 Alice Corp. v. CLS Bank International (573 U.S. 208, 2014)

Facts: Alice Corporation held patents on a computer-implemented method for mitigating settlement risk in financial transactions. The Federal Circuit initially found the claims patent-eligible; the Supreme Court reversed.

Holding: Abstract ideas implemented on a generic computer are not patent-eligible under 35 U.S.C. § 101. The Court articulated a two-step framework: (i) determine whether the claims at issue are "directed to" a patent-ineligible concept (law of nature, natural phenomenon, abstract idea); (ii) if so, search for an "inventive concept" transforming the claim into a patent-eligible application.

Implications: The Alice decision dramatically reshaped software patenting in the United States, invalidating broad business-method claims. The USPTO has issued multiple guidance revisions to apply Alice consistently, the latest reflecting developments through 2024.

7.2 Novartis AG v. Union of India (2013, Delhi High Court)

Facts: Novartis sought patent protection for the beta-crystallographic form of imatinib mesylate (Gleevec/Glivec). The application was rejected under Section 3(d) of the Indian Patents Act, which bars the patenting of new forms of known substances unless they exhibit significantly enhanced efficacy.

Holding: The Supreme Court of India (Novartis AG v. Union of India, Civil Appeal No. 2716 of 2013) upheld rejection, holding that the increased bioavailability, while therapeutically meaningful, did not satisfy the enhanced efficacy threshold.

Implications: The judgment illustrates how national implementation of TRIPS flexibilities can prioritize public health over pharmaceutical patenting, a topic directly relevant for India-based software firms working in health-tech.

7.3 Tesla Motors Patents and Open-Source Declarations (2014)

Facts: In June 2014, Tesla Motors announced via a blog post that it would not initiate patent infringement suits against any party using its electric vehicle technology in good faith.

Analysis: While Tesla's pledge was widely characterized as opening its patents, legally it was a conditional non-assertion pledge, not a dedication to the public domain. This case highlights how patent administration decisions (licensing, pledges, FRAND commitments) shape competitive landscapes; in software, equivalent moves include **Open Invention Network** participation or **patent non-assertion pledges**.

7.4 EPO and AI-Inventorship (Thaler v. President of the EPO, Case J 8/20, 2021)

Facts: Stephen Thaler's "DABUS" system (a Creativity Machine) was named as inventor on two patent applications.

Holding: The EPO Legal Board of Appeal held that under the European Patent Convention, the inventor must be a natural person, with legal personality attributed to the applicant. AI cannot be designated as an inventor.

Implications: With AI-generated inventions increasing in computer science, the patent system's administrative subject — the human inventor — has become a high-stakes legal question, with knock-on effects on ownership, inventorship, and remuneration in research institutions.

7.5 Indian Computer-Related Inventions — Eastern Tobacco Products v. Controller General (Delhi High Court, 2017)

Facts: A writ challenging the 2016 CRI Guidelines of the Indian Patent Office was filed on grounds that the Guidelines exceeded the scope of Section 3(k) of the Patents Act.

Holding: The Court declined to quash the Guidelines, observing that they merely restated the legal position and that computer-related inventions are examinable through the technical-effect analysis.

Implications: The case illustrates judicial deference to administrative interpretation and clarifies how India administers Section 3(k), a frequent flashpoint in software patenting.

8. Linkage with IPR in Educational Institutes

Although the present lecture focuses on international and administrative dimensions, the Unit 6 syllabus also highlights **IPR in educational institutes**. Briefly, the administration of patents directly affects universities through:

- **Bayh-Dole-style regimes** (e.g., India's Section 10 of the Patents Act allowing ownership of inventions arising from academic research).
- **Technology Transfer Offices (TTOs)** — a parallel administrative tier negotiating licensing of academic inventions.
- **Educational use exemptions** available in some copyright contexts but absent in patent law.
- **Researcher obligations** under funding agreements (e.g., DST, SERB, and industry-sponsored grants) requiring disclosure of inventions.

These considerations reinforce the practical relevance of patent administration for computer science researchers operating in academic settings.

9. Summary

- The **administration of the patent system** operates through national, regional, and international bodies, with **WIPO** coordinating treaty administration, **TRIPS** setting minimum standards within WTO law, and the **PCT** providing an international filing and search framework.
 - **Recent reforms** emphasize prosecution efficiency (Track One, PPH), accommodation of software inventions (USPTO § 101 guidance, EPO Compendium, Indian CRI Guidelines), and digital modernization accelerated by COVID-19.
 - **TRIPS** establishes the floor of obligations (20-year term, non-discrimination across technology fields) while permitting flexibilities (compulsory licensing, Bolar exemptions, parallel imports). TRIPS-plus bilateral obligations have layered additional commitments on many states.
 - The **PCT** does not grant international patents but provides a 30/31-month deferral and high-quality prior art search, materially benefiting computer science innovation.
 - **Case studies** (Alice, Novartis, DABUS/Thaler, Tesla, Eastern Tobacco) illustrate how administrative interpretation, treaty design, and judicial reasoning intertwine to shape patent practice.
 - For computer science researchers, fluent engagement with patent administration is now a core research-skill competency, supporting freedom-to-operate analysis, technology transfer, and strategic filing decisions.
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10. Exam Preparation Tips and Practice Questions

10.1 Tips for Answering Exam Questions on This Topic

1. **Distinguish tiers clearly.** Always separate national patent offices (USPTO, IPO), regional bodies (EPO, ARIPO, OAPI, EAPO), and international organizations (WIPO, WTO). Examiners reward precise institutional attribution.
2. **Use Articles, Sections, and Dates.** Citing TRIPS Article 27, PCT Article 8, Section 3(k) of the Indian Patents Act, or the AIA 2011 illustrates command of the textual framework.
3. **Anchor arguments in cases.** For software subject matter, cite *Alice v. CLS Bank*. For TRIPS flexibilities, cite *Novartis v. Union of India*. For AI inventorship, cite *J 8/20*.
4. **Compute procedural timelines.** Questions testing PCT procedure often expect 18-month publication and 30/31-month national-phase deadlines.

5. **Connect to TRIPS flexibilities.** When asked about pharmaceutical or software patenting, mention the trade-off between Article 27(1) availability and Article 31/8 flexibilities.

10.2 Practice Questions

Short Answer (2–3 marks each)

1. List any four national patent offices and state their respective jurisdictions.
2. State the tenure of patent protection under TRIPS Article 33.
3. What is the role of WIPO in administering the PCT?
4. Distinguish between TRIPS and the Paris Convention.

Medium-Answer Questions (5 marks each)

5. Discuss the main provisions of TRIPS Article 27 and explain their significance for computer-related inventions.
6. Outline the procedural stages of a PCT international application and explain the strategic advantage of using the PCT for software products.
7. Discuss the major administrative reforms introduced by the USPTO after the **America Invents Act, 2011**, with particular emphasis on software-related subject matter.

Long-Answer / Essay Questions (10–15 marks each)

8. "TRIPS has harmonized patent administration globally but leaves significant flexibilities to member states." Critically analyze the statement with reference to Articles 7, 8, 27, and 31 of the TRIPS Agreement.
9. Compare and contrast the **administrative architecture** of the European Patent Office and the Indian Patent Office. Discuss how each administrative system treats computer-related inventions.
10. Critically examine the implications of the **DABUS / Thaler** line of decisions for computer science researchers relying on AI for inventive contributions. Discuss whether patent administration should evolve to recognize AI-generated inventions.
11. **Case-Study Based:** *A startup founded by B.Tech students at UPES develops a machine-learning algorithm that optimizes supply-chain logistics and seeks protection in India, the United States, and the European Union. Advise the startup on (a) how TRIPS applies in each jurisdiction, (b) the strategic use of the PCT, and (c) the probable patentability of the algorithm under current § 101 (US) and Section 3(k) (India) jurisprudence.*

Numerical/Quantitative Reasoning (Optional)

12. A PCT international application is filed with an Indian priority date of **1 January 2024**. Calculate the deadline for national-phase entry in India, the United States, and the European Union (assuming EPO designation). Justify the relevance of this deadline for the applicant.

10.3 Recommended Reading

- Bently, L., Sherman, B. (Eds.), *Intellectual Property Law* (Oxford University Press, latest edition).
- Correa, C. M., *Trade Related Aspects of Intellectual Property Rights: A Commentary on the TRIPS Agreement* (Oxford University Press).
- WIPO, *Background Reading Material on the Patent Cooperation Treaty* (latest revision).
- *Manual of the Examining Procedure (MPEP)* (USPTO, latest revision).
- *Guidelines for Examination in the European Patent Office (EPO)*.
- Indian Patent Office, *Manual of Patent Practice and Procedure and Computer-Related Inventions Examination Guidelines*.
- Singh, A., *Law Relating to Intellectual Property Rights* (LexisNexis).

Prepared for academic teaching within the Research Methodology in Computer Science course at UPES University. All statutory references and case citations should be verified against current official sources before use in assessment or publication.